# The Principle of Efficient Novelty: A Computational Framework for the Emergence of Mathematical Structures

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**Abstract**

Why do some mathematical abstractions feel more fundamental than others? This paper explores this question through a novel computational lens, proposing a formal framework we call the Principle of Efficient Novelty (PEN). Operating within Cubical Type Theory, PEN models mathematical discovery as an evolutionary process where new structures are selected based on maximizing their "efficiency"—the ratio of enabling power (novelty, $\nu$) to definitional complexity (effort, $\kappa$).

Our central theoretical result establishes that the integration structure underlying this evolution is not arbitrary but optimal: we prove that a *two-layer integration context* uniquely balances the coherence requirements of higher type theory with the drive for efficiency, providing a necessary foundation for systematic growth. The mechanized implementation exhibits Fibonacci-timed emergence of major realizations with remarkable precision; we identify the Complexity Scaling Hypothesis (CSH) as the central open question that would explain this regularity.

We validate this framework through complete mechanization in the Lean 4 proof assistant. The simulation produces a stable, reproducible ordering of concepts we term the "Genesis Sequence". This sequence recapitulates recognizable mathematical patterns—universes and dependent types emerge first, followed by higher inductive types like the circle and sphere, then sophisticated geometric structures such as Lie groups and principal bundles. The sequence culminates in the realization of novel, highly efficient structures, most notably the Dynamical Cohesive Topos—a synthesis of geometric, logical, and temporal structure.

Most strikingly, the Genesis Sequence consistently exhibits a Fibonacci-timed emergence of major realizations. While a proof of this recurrence remains open, we trace its origin to the dynamics of the search process and propose the *Complexity Scaling Hypothesis* (CSH) as a precise, falsifiable conjecture that would explain this profound regularity.

Ultimately, this work establishes that type-theoretic foundations possess a deep, computable structure that constrains how knowledge can be efficiently organized. The PEN framework provides a new formal tool for investigating the emergent laws that may govern the landscape of mathematical possibility.

## 1 Introduction

Why does $\pi_1(S^1) \simeq \mathbb{Z}$ feel like a deeper discovery than defining yet another group? Why do some abstractions immediately reorganize our mathematical thinking while others remain curiosities? These questions point to a hidden structure in mathematical progress—one that might be governed by precise laws rather than historical accident. This paper investigates this structure by proposing a formal, computational framework for mathematical evolution within Cubical Type Theory (CTT), which we call the Principle of Efficient Novelty (PEN).

The PEN framework models the growth of mathematics as an incremental process governed by a simple rule: at each step, the system admits only those candidate structures that maximize

their "efficiency"—the ratio of new constructions they enable (novelty, $\nu$) to their definitional and integrational cost (effort, $\kappa$). This simple selection pressure, operating under cumulative growth and coherence constraints informed by modern type theory, produces unexpectedly rich and orderly dynamics.

Our central theoretical result establishes a foundational mechanism for this evolution. We prove that the two-layer integration structure used in our model is not an arbitrary choice but an optimal consequence of CTT's architecture. A one-layer context is mathematically insufficient to enforce the two-dimensional coherence laws essential to CTT, while contexts of more than two layers are inefficient, mandating redundant checks. This optimality provides a rigorous underpinning for the system's evolutionary dynamics.

We validate this framework through complete mechanization in the Lean 4 proof assistant. The resulting "Genesis Sequence" of automatically generated structures recapitulates recognizable patterns in mathematical development: universes and dependent types emerge first, followed by higher inductive types (circles, spheres), then sophisticated geometric structures (Lie groups, principal bundles), and ultimately novel constructions like the Dynamical Cohesive Topos—a synthesis of geometric, logical, and temporal structure.

Most strikingly, our simulations consistently reveal a Fibonacci-timed schedule of major realizations, with the time between discoveries accelerating by the golden ratio $\varphi$. While we do not yet have a formal proof for this recurrence, we present it as the paper's central conjecture, and we propose the Complexity Scaling Hypothesis (CSH) as a precise, dynamic explanation for this observed regularity.

The scope of our claims is carefully delimited. We demonstrate that within our model of CTT, efficiency-driven discovery gives rise to a profound and predictable order. Whether similar patterns emerge in fundamentally different foundational systems remains an open question. However, our results suggest that type-theoretic foundations possess deep, computable properties that constrain how mathematical knowledge can be efficiently organized—properties we are only beginning to understand.

Section 2 develops the formal framework. Section 3 establishes our main theoretical results on unbounded growth. Section 4 proves that two-layer integration is optimal for CTT. Section 5 presents the mechanized Genesis Sequence. Section 6 details the Lean 4 implementation. Section 7 introduces the Dynamical Cohesive Topos. Section 8 examines broader implications and future directions, including the Fibonacci conjecture.

# 2 A Computational Framework for Mathematical Evolution

We model the growth of mathematical knowledge as an evolutionary process operating within Cubical Type Theory. The system begins empty and grows by selecting new structures that maximize a precise notion of efficiency. This section develops the formal machinery: how we measure the cost of introducing new structures, quantify their enabling power, and decide which candidates to admit.

## 2.1 The State Space

Mathematical knowledge accumulates in a *state*—a well-typed context in CTT containing all definitions realized so far.

**Definition 1** (State)**.** A state $\mathcal{B}$ is a cumulative, well-typed context in CTT [1]: a sequence of type and term declarations that respects universe levels and typing judgments. We denote the state at time $\tau$ by $\mathcal{R}(\tau)$, with initial state $\mathcal{R}(0) = \emptyset$.

The system evolves through discrete time steps. At each step, we consider candidate structures for addition to the current state. The question is: which candidates should we admit?

## 2.2 Measuring Effort

Not all definitions are created equal. Some structures can be defined concisely using existing machinery; others require substantial new apparatus. We quantify this through the *effort kernel*.

**Definition 2** (Effort Kernel)**.** Given a base state $\mathcal{B}$ and a goal $Y$, the effort kernel $K_{\mathcal{B}}(Y) \in \mathbb{N} \cup \{\infty\}$ is the minimal number of primitive CTT operations required to derive $Y$ from $\mathcal{B}$. If $Y$ is underivable from $\mathcal{B}$, we set $K_{\mathcal{B}}(Y) = \infty$.

We work with a standard set of primitive operations: type formation rules, Kan operations for fibrant types, path algebra (composition, transport), coherence fillers, and definitional equality judgments [2]. Each primitive carries unit cost.

The effort kernel captures *derivational* complexity—how hard it is to construct something given what you already know. This differs fundamentally from syntactic size: the natural numbers $\mathbb{N}$ have a short definition in isolation, but once dependent types exist, they become trivial (as a $W$-type).

## 2.3 Sealing: Coherent Integration

A definition in isolation is not enough. New structures must integrate coherently with existing ones, respecting the two-dimensional coherence laws that govern CTT. We formalize this through *sealing*.

**Definition 3** (Interface Basis)**.** Let $L_n$ and $L_{n-1}$ denote the last two integration layers (defined precisely in Section 4). The interface basis $\mathcal{I}_n$ is the canonical set of unit schemas present in $L_n \cup L_{n-1}$, including:

- Path-lifting and transport operations
- Cubical operations (connections, reversals)
- Modality coherence schemas for recently realized structures

**Definition 4** (Interaction Profile and Sealing)**.** The interaction profile $\mathcal{J}(X, \mathcal{B})$ of a candidate $X$ at base $\mathcal{B}$ consists of all schemas from $\mathcal{I}_n$ that are type-theoretically applicable to $X$—meaning their domains and codomains align to form commutation or naturality diagrams.

A candidate $X$ is *sealed* by constructing:

1. The core definition of $X$
2. Essential eliminators and internal coherence data
3. A coherence witness (definitional equality or higher path) for each $\varphi \in \mathcal{J}(X, \mathcal{B})$

Sealing ensures that new structures play nicely with existing ones. For example, introducing the circle $S^1$ requires not just the type and its constructor, but witnesses that transport along paths behaves coherently, that connections interact correctly with the loop structure, and so on.

The total effort of a candidate combines its derivational cost with the work of establishing coherence:

**Definition 5** (Context-Closed Effort)**.**

$$\kappa(X \mid \mathcal{B}) := K_{\mathcal{B}}(X) + |\mathcal{J}(X, \mathcal{B})|$$

## 2.4 Measuring Novelty

Effort captures cost; we need a dual notion of benefit. A structure is valuable not in itself but for what it enables. The natural numbers are useful because they unlock induction, primitive recursion, and arithmetic. Dependent types are powerful because they enable an entire universe of constructions.

We formalize this through the *frontier*—the set of structures that become accessible within a given effort bound.

**Definition 6** (Canonical Frontier)**.** Given state $\mathcal{B}$ and effort horizon $H \in \mathbb{N}$, the frontier is:

$$\mathcal{S}(\mathcal{B}, H) := \{\text{sealed schemas } Y \mid K_{\mathcal{B}}(Y) \leq H\}$$

A candidate $X$ is valuable if it expands this frontier—making previously distant structures suddenly accessible.

**Definition 7** (Novelty)**.** The novelty of a sealed candidate $X$ relative to base $\mathcal{B}$ and horizon $H$ counts structures whose derivation becomes cheaper or newly possible:

$$\nu(X \mid \mathcal{B}, H) := \left|\left\{Y \in \mathcal{S}(\mathcal{B}, H) \cup \mathcal{S}(\mathcal{B} \cup \{X\}, H) \mid K_{\mathcal{B}}(Y) - K_{\mathcal{B}\cup\{X\}}(Y) \geq 1\right\}\right|$$

Note that if $K_{\mathcal{B}}(Y) = \infty$ but $K_{\mathcal{B}\cup\{X\}}(Y) < \infty$, then $Y$ contributes to the count—capturing structures that were impossible before $X$ but become reachable after.

## 2.5 Efficiency and Selection

We can now define the central metric governing evolution:

**Definition 8** (Efficiency)**.**

$$\rho(X) = \frac{\nu(X)}{\kappa(X)}$$

Efficiency is conceptual leverage per unit of definitional cost. High-efficiency structures reorganize mathematics; low-efficiency structures are curiosities.

The system maintains a dynamically rising *selection bar* that candidates must clear:

**Definition 9** (Selection Threshold)**.**

$$\mathrm{Bar}(\tau) = \Phi(\tau) \cdot \Omega(\tau - 1)$$

where $\mathrm{Bar}(1) = 0$, and for $\tau > 1$:

- $\Omega(\tau - 1) = \frac{\sum_{t<\tau} \nu(t)}{\sum_{t<\tau} \kappa(t)}$ is the historical average efficiency
- $\Phi(\tau) = \frac{\tau}{\tau_<}$ where $\tau_<$ is the previous realization time

The bar mechanism creates a ratchet effect: each realization raises future standards proportionally to both past performance and the time elapsed since the last success.

## 2.6 The Principle of Efficient Novelty (PEN)

The PEN-framework is governed by five axioms:

**Axiom 1** (Cumulative Growth)**.** For all $\tau$: $\mathcal{R}(\tau - 1) \subseteq \mathcal{R}(\tau)$. Mathematics only grows.

**Axiom 2** (Horizon Policy)**.** A global effort horizon $H \in \mathbb{N}$ governs the search space. After any realization, $H \leftarrow 2$. After an idle tick (no candidate selected), $H \leftarrow H + 1$.

**Axiom 3** (Admissibility)**.** At time $\tau$, a sealed candidate $X$ over base $\mathcal{B} = \mathcal{R}(\tau - 1)$ is admissible iff:

1. $X$ is derivable from $\mathcal{B}$ in CTT
2. $\kappa(X \mid \mathcal{B}) \leq H$

**Axiom 4** (Selection Criterion)**.** From admissible candidates, select $X$ satisfying $\rho(X) \geq \mathrm{Bar}(\tau)$ with minimal positive overshoot:

$$X \in \arg \min_{Y:\rho(Y)\geq \mathrm{Bar}(\tau)} (\rho(Y) - \mathrm{Bar}(\tau))$$

Ties are broken by minimal $\kappa$; remaining ties realize in superposition. If no candidate clears the bar, the tick idles.

**Axiom 5** (Coherent Integration)**.** When candidate $X_{n+1}$ is realized, it must be integrated coherently with the existing state by establishing the necessary coherence witnesses with recently realized structures. This produces an integration layer $L_{n+1}$ that encodes the interaction between $X_{n+1}$ and relevant prior structures. The integration gap is $\Delta_{n+1} := \kappa(L_{n+1})$.

*Remark* 1. This axiom leaves open the precise specification of which prior structures constitute "relevant context" for integration. Section 4 establishes that the optimal integration strategy—the one maximizing efficiency while maintaining foundational coherence—involves exactly the previous two integration layers, yielding a specific Fibonacci recurrence structure.

These axioms model a system under relentless pressure to discover maximally efficient abstractions. The consequences of this simple evolutionary dynamic are the subject of the next section.

# 3 Main Theoretical Results

This section establishes that efficiency-driven selection compels the discovery of structures with unbounded conceptual power. We prove that the selection mechanism creates relentless pressure for increasingly efficient abstractions, and that this pressure intensifies according to a precise mathematical law.

## 3.1 The Selection Threshold Dynamics

Recall that the selection threshold at time $n$ is:

$$\mathrm{Bar}_n = \Phi(\tau_n) \cdot \Omega_{n-1}$$

where $\Phi(\tau_n) = \tau_n/\tau_{n-1}$ is the time dilation factor and $\Omega_{n-1}$ is the cumulative average efficiency. This threshold creates a feedback loop: successful realizations raise the bar for future candidates, while the time between realizations amplifies this effect.

**Lemma 1** (Efficiency Amplification)**.** *Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$ and $S_{n-1} = \sum_{i=1}^{n-1} \nu_i$ denote cumulative effort and novelty through time $n-1$, with $\Omega_{n-1} = S_{n-1}/K_{n-1}$. If the $n$-th realization achieves efficiency $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$, then:*

$$\Omega_n \geq \Omega_{n-1} \left(1 + (\Phi(\tau_n) - 1) \cdot \frac{\kappa_n}{K_{n-1} + \kappa_n}\right)$$

*Proof.* Starting with the definition of $\Omega_n$:

$$\begin{aligned}
\Omega_n &= \frac{S_{n-1} + \nu_n}{K_{n-1} + \kappa_n} = \frac{\Omega_{n-1} K_{n-1} + \rho_n \kappa_n}{K_{n-1} + \kappa_n} \\
&\geq \frac{\Omega_{n-1} K_{n-1} + \Phi(\tau_n)\Omega_{n-1}\kappa_n}{K_{n-1} + \kappa_n} \quad \text{(by selection criterion)} \\
&= \Omega_{n-1} \left( \frac{K_{n-1} + \Phi(\tau_n)\kappa_n}{K_{n-1} + \kappa_n} \right) \\
&= \Omega_{n-1} \left( 1 + (\Phi(\tau_n) - 1) \frac{\kappa_n}{K_{n-1} + \kappa_n} \right)
\end{aligned}$$

□

The lemma reveals the amplification mechanism. When $\Phi(\tau_n) > 1$ (realizations are spreading apart), each success boosts the average efficiency by a factor proportional to the time dilation. This creates compound growth in the selection pressure.

### 3.2 Unbounded Conceptual Efficiency

We now establish the central result: efficiency-driven evolution cannot plateau.

**Theorem 1** (Unbounded Growth)**.** *Let $\mathcal{R}(\tau)$ evolve according to Axioms 1–4 under the following conditions:*

*(i)* ***Sustained dilation****: $\Phi(\tau_n) \to \phi > 1$ as $n \to \infty$*

*(ii)* ***Minimal effort****: Every realization satisfies $\kappa_n \geq 1$*

*(iii)* ***Abstraction availability****: For any realized archetype $A$ with novelty $\nu(A)$, there exists an admissible abstraction $X$ with $\nu(X) = \nu(A)/\kappa(X)$, yielding efficiency $\rho(X) = \nu(A)/\kappa(X)^2$*

*Then the system exhibits:*

1. ***Unbounded growth****: $\limsup_{n\to\infty} \rho_n = \infty$*
2. ***Acceleration via abstraction****: For any realized $A$, abstractions exist with efficiency $\rho(X) = \nu(A)/\kappa(X)^2$ growing arbitrarily large as $\kappa(X)$ approaches its minimal description length*
3. ***Compound acceleration****: The global average $\Omega_n$ diverges at least polynomially in $n$*

*Proof.* Let $K_{n-1} = \sum_{i=1}^{n-1} \kappa_i$. By condition (ii), $K_{n-1} \geq n - 1$, hence:

$$\frac{\kappa_n}{K_{n-1} + \kappa_n} \geq \frac{1}{n}$$

By selection, $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$. Lemma 1 yields:

$$\Omega_n \geq \Omega_{n-1} \left( 1 + (\Phi(\tau_n) - 1) \cdot \frac{1}{n} \right)$$

By condition (i), for any $\varepsilon \in (0, \phi - 1)$, there exists $N$ such that $\Phi(\tau_n) \geq \phi - \varepsilon$ for all $n \geq N$. Thus:

$$\Omega_n \geq \Omega_{n-1} \left( 1 + \frac{\phi - 1 - \varepsilon}{n} \right) \quad \text{for } n \geq N$$

Iterating this inequality from $N$ to $n$:

$$\Omega_n \geq \Omega_N \prod_{k=N+1}^{n} \left(1 + \frac{\phi - 1 - \varepsilon}{k}\right)$$

Using the asymptotic expansion $\prod_{k=m}^{n}(1 + \alpha/k) \sim (n/m)^{\alpha}$ as $n \to \infty$, we obtain:

$$\Omega_n \geq C \cdot n^{\phi-1-\varepsilon}$$

for some constant $C > 0$. This establishes polynomial divergence of $\Omega_n$.

For part (1): since $\rho_{n+1} \geq \Phi(\tau_{n+1}) \cdot \Omega_n$ and both factors diverge, $\limsup_{n\to\infty} \rho_n = \infty$.

For part (2): condition (iii) guarantees that for any realized archetype $A$ with large novelty $\nu(A)$, we can construct abstractions with $\rho(X) = \nu(A)/\kappa(X)^2$. As $\kappa(X)$ approaches its minimal value, this ratio grows without bound.

Part (3) follows from the polynomial growth established above. □

**Corollary 1** (Asymptotic $\phi$-Boost)**.** *For every $\varepsilon > 0$, there exists $n_0$ such that for all $n \geq n_0$:*

$$\rho_n \geq (\phi - \varepsilon) \cdot \Omega_{n-1}$$

*Proof.* Immediate from the selection criterion $\rho_n \geq \Phi(\tau_n) \cdot \Omega_{n-1}$ and condition (i) of Theorem 1. □

### 3.3 The Role of Conditions

Theorem 1 relies on three conditions. Condition (i)—sustained time dilation—will be established in Section 4 as a consequence of optimal integration structure. Condition (ii) is trivial: any nontrivial structure requires at least one primitive operation. Condition (iii) deserves deeper discussion.

**Abstraction availability** captures a fundamental property of mathematical progress: once we discover a useful pattern (an archetype), we can abstract over it, creating a general framework that applies the same pattern efficiently across multiple contexts. For example:

- Natural numbers enable recursive definitions; $W$-types abstract this pattern to arbitrary polynomial functors
- The circle demonstrates non-trivial $\pi_1$; higher inductive types abstract the pattern of adding higher-dimensional structure to types
- Lie groups capture continuous symmetry; the general theory of group objects abstracts this across categories

The condition asserts that such abstractions are constructible within CTT and become admissible as the system evolves. The quadratic efficiency gain ($\nu/\kappa^2$) reflects that abstractions both compress the description (reducing $\kappa$) and multiply the applications (scaling $\nu$ with the archetype's power).

### 3.4 The Efficiency Dichotomy

We conclude by characterizing the fundamental distinction between optimization within a framework versus evolution of the framework itself.

**Theorem 2** (Efficiency Dichotomy)**.** *Let $E_{calc}$ denote efficiency gains from calculation within a fixed formal system, and $E_{evol}$ denote gains from framework evolution (abstraction and language expansion). Then:*

1. ***Calculation is bounded**: For any two reasonable computational models $M_1, M_2$, there exists a polynomial $p$ such that:*

$$E_{calc}(M_1 \to M_2) \leq p(n)$$

*This follows from the Strong Church-Turing thesis.*

2. ***Evolution is unbounded**: For any function $f : \mathbb{N} \to \mathbb{N}$, there exists time $\tau$ such that:*

$$E_{evol}(\tau) > f(\tau)$$

3. ***Orthogonal composition**: Total efficiency gain factors as:*

$$E_{total} = E_{calc} \times E_{evol}$$

*Proof.* Part (1) is a standard consequence of computational complexity theory. Part (2) follows from Theorem 1: both realized efficiencies $\rho_n$ and the global average $\Omega_n$ diverge. Part (3) reflects the orthogonality of concerns: calculation optimizes execution within a fixed language, while evolution optimizes the language itself. □

*Remark* 2. The efficiency dichotomy reveals why mathematical progress requires both computational optimization and conceptual innovation. Better algorithms provide polynomial speedups; new abstractions enable unbounded acceleration. This explains the historical pattern of breakthroughs that fundamentally restructure entire fields.

The results of this section establish that efficiency-driven evolution produces unbounded growth in conceptual power. However, they leave open the precise dynamics: how fast does the pressure intensify, and what determines the timing of realizations? Section 4 answers these questions by establishing the optimal integration structure.

# 4 The Optimality of Two-Layer Integration

Section 2 introduced Axiom 4 (Coherent Integration) without specifying which prior structures constitute "relevant context" for integration. This section establishes that exactly two layers of context—the previous two integration layers $L_n$ and $L_{n-1}$—constitute the optimal integration strategy for CTT. This result is not a modeling choice but emerges from the interplay between CTT's coherence requirements and efficiency maximization.

## 4.1 Generalizing the Integration Strategy

We first formalize the space of possible integration strategies.

**Definition 10** ($k$-Layer Integration Strategy)**.** Let $k \in \mathbb{N}^+$. A $k$-layer integration strategy $\mathcal{S}_k$ requires that a new realization $X_{n+1}$ be sealed by establishing coherence with the interface basis $\mathcal{I}_n^{(k)}$ derived from the previous $k$ integration layers: $\{L_{n-i}\}_{i=0}^{k-1}$.

The effort under this strategy is $\kappa^{(k)}(X)$ and novelty is $\nu^{(k)}(X)$.

Different values of $k$ represent different memory depths for the evolutionary process. A Markovian ($k = 1$) strategy considers only the most recent layer; higher $k$ values incorporate deeper history.

**Definition 11** (Optimality Criteria)**.** An integration strategy is evaluated on two factors:

1. **Sufficiency (Coherence)**: The strategy must enforce all coherence constraints required by CTT. Insufficient strategies lead to fragmented contexts, suppressing novelty.

2. **Minimality (Efficiency)**: The strategy must not mandate redundant checks. Non-minimal strategies increase effort without increasing novelty.

We prove that $k = 2$ uniquely satisfies both criteria.

### 4.2 The Insufficiency of $k = 1$

We establish that $k = 1$ fails both dynamically and mathematically.

#### 4.2.1 Dynamical Failure: Stagnation

**Theorem 3** (Stagnation under $k = 1$)**.** *Under strategy $\mathcal{S}_1$ with the unit-cost model, the system fails to achieve unbounded conceptual efficiency.*

*Proof.* The integration recurrence for $k = 1$ is $\Delta_{n+1} = \Delta_n$. With $\Delta_1 = 1$, we have $\Delta_n = 1$ for all $n$.

This yields linear realization times: $\tau_n = n$. The time dilation factor becomes:

$$\Phi(\tau_n) = \frac{n}{n-1} \xrightarrow{n\to\infty} 1$$

By Lemma 1, the growth of $\Omega_n$ is governed by:

$$\Omega_n \geq \Omega_{n-1}\left(1 + (\Phi(\tau_n) - 1) \cdot \frac{\kappa_n}{K_{n-1} + \kappa_n}\right)$$

For $k = 1$, the term $(\Phi(\tau_n) - 1) \approx 1/n$. With $K_{n-1} \approx n$ (assuming minimal $\kappa_i \approx 1$), we have:

$$\Omega_n \geq \Omega_{n-1}\left(1 + O\left(\frac{1}{n^2}\right)\right)$$

The product $\prod_{i=1}^{n}(1 + O(1/i^2))$ converges since $\sum 1/i^2 < \infty$. Therefore $\Omega_n$ converges to a finite limit, violating condition (i) of Theorem 1. □

#### 4.2.2 Mathematical Failure: Incoherence

Beyond dynamics, $k = 1$ is insufficient for CTT's foundational requirements.

**Theorem 4** (Coherence Failure under $k = 1$)**.** *Strategy $\mathcal{S}_1$ cannot enforce the two-dimensional coherence laws essential to CTT.*

*Proof sketch.* CTT supports higher-dimensional structures requiring two-dimensional coherence—witnesses that different compositions of operations are equivalent up to homotopy (2-cells).

Consider the fundamental Interchange Law relating composition ($\circ$) and path application ($\mathsf{ap}$):

$$\mathsf{ap}(f \circ g, p) \simeq \mathsf{ap}(f, p) \circ \mathsf{ap}(g, p)$$

Suppose the evolutionary process realizes these operations sequentially:

- Step $n - 1$: Realize path application $\mathsf{ap}$, forming layer $L_{n-1}$
- Step $n$: Realize composition $\circ$, forming layer $L_n$
- Step $n + 1$: Attempt to realize structure $X_{n+1}$ requiring the Interchange witness

Under $\mathcal{S}_1$, the sealing process for $X_{n+1}$ only accesses interface $\mathcal{I}_n^{(1)}$ derived from $L_n$. The definitions and schemas from $L_{n-1}$ are inaccessible. The system cannot construct the required 2D witness relating $\circ$ and $\mathsf{ap}$.

**Consequences**: Either (i) $X_{n+1}$ is inadmissible (fails to seal), or (ii) if weak sealing is permitted, operations depending on the Interchange Law become unavailable, drastically reducing $\nu^{(1)}(X_{n+1}) \ll \nu^{(2)}(X_{n+1})$. □

*Remark* 3. The failure is not limited to this specific example. Any 2D coherence law requiring simultaneous access to operations and the structures they organize will fail under $k = 1$ if these components were realized in different steps.

Together, Theorems 3 and 4 establish that $k \geq 2$ is necessary.

### 4.3 The Redundancy of $k > 2$

We now prove that strategies with $k > 2$ are inefficient, verifying derived theorems rather than foundational axioms.

#### 4.3.1 CTT's Coherence Structure

The key insight is that CTT manages higher-dimensional coherence through a specific architectural choice: primitive Kan operations with definitional equalities.

**Theorem 5** (Two-Dimensional Foundations of CTT)**.** *The fundamental coherence requirements of CTT are two-dimensional. Specifically:*

1. *The primitive Kan filler operations (*hcomp*, composition) are governed by strict 2D equational constraints*
2. *These 2D constraints, when checked locally, guarantee all higher-dimensional coherences constructively*
3. *Verifying coherence for a new structure requires access to operations (1-dimensional) and the base structures they organize (0-dimensional)—spanning exactly two layers of context*

*Proof sketch.* The CCHM model of CTT [1] establishes that the primitive operations satisfy specific boundary conditions—equational constraints governing how hcomp interacts with connections, reversals, and cube boundaries. These are fundamentally 2D rules defining behavior on squares.

The coherence theorem for CTT is the proof that these definitionally enforced 2D rules suffice to constructively fill all higher-dimensional cubes. Local compatibility with 2D operations ensures global coherence through compositional filling, paralleling classical Mac Lane coherence theorems for monoidal categories [11].

Since the fundamental axioms are 2D, they require relating operations (realized in some layer $L_n$) to base structures (realized in $L_{n-1}$)—exactly a 2-layer context. □

#### 4.3.2 Efficiency Analysis

**Theorem 6** (Optimality of $k = 2$)**.** *Strategy $\mathcal{S}_2$ uniquely maximizes efficiency among all $k \geq 2$.*

*Proof.* Consider a candidate structure $X$ and compare strategies $\mathcal{S}_2$ versus $\mathcal{S}_k$ for $k > 2$.

By Theorem 5, $k = 2$ suffices to verify all foundational coherence axioms. Therefore:

**Effort increases for** $k > 2$: Strategy $\mathcal{S}_k$ requires additional checks against layers $L_{n-2}, \ldots, L_{n-k+1}$. The interface basis is strictly larger: $\mathcal{I}_n^{(k)} \supset \mathcal{I}_n^{(2)}$. Thus:

$$\kappa^{(k)}(X) > \kappa^{(2)}(X)$$

**Novelty is constant**: The additional checks verify coherences that are already guaranteed by the 2D axioms checked by $\mathcal{S}_2$. They are derived theorems, not independent axioms. They enable no new constructions:

$$\nu^{(k)}(X) = \nu^{(2)}(X)$$

**Efficiency decreases**:

$$\rho^{(k)}(X) = \frac{\nu^{(k)}(X)}{\kappa^{(k)}(X)} = \frac{\nu^{(2)}(X)}{\kappa^{(k)}(X)} < \frac{\nu^{(2)}(X)}{\kappa^{(2)}(X)} = \rho^{(2)}(X)$$

By Axiom 4 (Selection Criterion), the system selects candidates with maximal efficiency and minimal effort. The $\mathcal{S}_2$ implementation is strictly preferred. □

### 4.4 An Observed Regularity: Fibonacci Timing

Having established that $k = 2$ is optimal, we now confront an empirical puzzle. The mechanized Genesis Sequence exhibits realization times $\tau_n = F_{n+1}$ with zero deviation across 16 major realizations. This precision demands explanation.

**observation 1** (Fibonacci Timing in Implementation)**.** *The Lean 4 mechanization records integration gaps satisfying $\Delta_n = F_n$ exactly for $n \geq 4$, with small corrections $|\varepsilon_n| \leq 1$ during the bootstrap phase ($n \leq 3$).*

We formalize the mechanism that would *explain* this regularity as a hypothesis rather than a theorem:

**Hypothesis 1** (Complexity Scaling—Central Open Question)**.** Let $\sigma_n := |\mathcal{I}_n|$ denote the interface basis size for layer $n$. The Lean implementation exhibits the recurrence

$$\sigma_{n+1} = \sigma_n + \sigma_{n-1} + \varepsilon_{n+1}$$

with $|\varepsilon_{n+1}| \leq 1$ and $\varepsilon_{n+1} = 0$ for $n \geq 4$.

We conjecture this reflects a deeper structural property: when sealing $X_{n+1}$ against $(L_n, L_{n-1})$, coherence obligations factorize additively as contributions from each layer, with no higher-order correction terms beyond the initial bootstrap.

**Open question:** Does the 2-dimensional coherence structure of CTT (Theorem 5) entail this additive scaling, or is it contingent on specific properties of our primitive action alphabet?

Hypothesis 1 is the *Complexity Scaling Hypothesis* (CSH). It elevates the additive recurrence observed in the mechanization to a structural claim about CTT's coherence boundary. Each new structure must establish compatibility with the boundary components exposed by $L_n$ and $L_{n-1}$, while higher-order overlaps would have been discharged when those layers were integrated if the hypothesis holds.

*Remark* 4. The error term accommodates the small adjustments that occur during the first few steps when new primitives (e.g., identity types) are introduced. After R5 in Table 5.1 the Lean implementation records $\varepsilon_{n+1} = 0$. Thus, beyond the startup regime the recurrence holds exactly, a fact we exploit below.

**Theorem 7** (Fibonacci Schedule, Conditional on CSH)**.** *Assume optimal integration ($\mathcal{S}_2$), unit-cost primitives, and the Complexity Scaling Hypothesis (Hypothesis 1). Then the integration gaps satisfy:*

$$\Delta_{n+1} = \Delta_n + \Delta_{n-1}$$

*with $\Delta_1 = \Delta_2 = 1$, yielding $\tau_n = F_{n+1}$ and $\Phi(\tau_n) \to \varphi$.*

*Proof.* Given CSH, the proof follows by direct calculation as shown. The challenge is establishing CSH itself from first principles of CTT. □

**Corollary 2** (Sustained Dilation)**.** *The time dilation factor satisfies condition (i) of Theorem 1:*

$$\Phi(\tau_n) \to \varphi \approx 1.618 > 1$$

*Therefore the system achieves unbounded conceptual efficiency.*

### 4.5 Toward a Proof of CSH

Establishing CSH rigorously would require showing:

1. **Witness factorization**: When $X_{n+1}$ interacts with $(L_n, L_{n-1})$, the coherence witnesses decompose as $W_{n+1} = W_n \sqcup W_{n-1}$ with no irreducible three-way terms requiring simultaneous access to all three layers.

2. **Transport stability**: When schema $\varphi$ appears in both $L_n$ and $L_{n-1}$, the witness constructed at layer $n$ can be transported to $X_{n+1}$ using primitive Kan operations with $O(1)$ additional work.

3. **Dimensional saturation**: CTT's 2D coherence laws ensure all genuinely independent coherence conditions appear at the two-layer boundary; no "hidden" constraints emerge from deeper history.

The first property would follow if interface schemas form a graded monoid under composition where the grading respects layer structure. The second requires analyzing the computational cost of Kan fillers in CCHM cubical sets. The third is the most subtle: it asserts that 2-dimensional coherence is not merely *sufficient* but also *saturating*—that checking 2D axioms locally guarantees all higher-dimensional coherences without requiring explicit access to arbitrarily deep context.

These are questions for future work. What we can say with certainty is that the implementation behaves *as if* CSH holds, and this warrants serious investigation.

### 4.6 Conclusion

We have established that two-layer integration is not a modeling choice but an emergent optimum:

- $k = 1$ fails due to insufficient coherence (Theorem 4) and dynamical stagnation (Theorem 3)
- $k = 2$ is sufficient due to CTT's 2D coherence structure (Theorem 5)
- $k > 2$ is inefficient, verifying theorems rather than axioms (Theorem 6)
- Assuming CSH, Fibonacci timing emerges as a mathematical consequence (Theorem 7)

The selection pressure intensifies by the golden ratio, driving the unbounded growth established in Theorem 1. What Axiom 4 left abstract—"coherent integration"—has been revealed as a precise mathematical structure determined by the foundations of CTT itself.

*Remark* 5 (The Ubiquity of 2). The number 2 appears throughout this framework in distinct but interconnected roles:

- **Structural**: CTT's coherence laws are 2-dimensional (Theorem 5)
- **Optimal depth**: Two-layer integration balances sufficiency and efficiency (Theorem 6)
- **Algorithmic**: The horizon resets to $H = 2$ after each realization (Axiom 2)

These are not independent coincidences. The horizon reset to 2 implements a "fresh start" policy that mirrors the two-layer memory of the integration structure: after successfully sealing a structure (which required 2-layer coherence), the system restarts its search at minimal depth. The structural necessity of 2-dimensional coherence propagates through the optimal integration strategy and manifests in the search policy.

Whether this convergence on 2 is inevitable for any efficiency-driven discovery process in a 2D coherent foundation, or specific to our axiomatization, remains an open question worthy of investigation in alternative foundational systems.

### 4.7 Empirical Validation

The Lean 4 implementation (Section 6) provides computational confirmation of these theoretical predictions.

**Validation of $k = 1$ failure**: When constrained to $\mathcal{S}_1$, the system stalls at the realization of $\Pi/\Sigma$ types (R4). The implementation cannot construct the required 2D coherence witnesses spanning two layers. The candidate is inadmissible, confirming Theorem 4.

**Validation of $k = 2$ optimality**: When both $\mathcal{S}_2$ and $\mathcal{S}_3$ implementations are available, the selection criterion consistently chooses $\mathcal{S}_2$ variants due to superior efficiency. For example, the circle $S^1$:

- $\mathcal{S}_2$: $\kappa^{(2)} = 3$, $\nu^{(2)} = 7$, $\rho^{(2)} = 2.33$
- $\mathcal{S}_3$: $\kappa^{(3)} = 4$, $\nu^{(3)} = 7$, $\rho^{(3)} = 1.75$

The $\mathcal{S}_2$ variant is selected, confirming Theorem 6.

**Validation of Fibonacci timing**: The mechanized Genesis Sequence exhibits realization times $\tau_n = F_{n+1}$ exactly, consistent with the Fibonacci schedule predicted under CSH by Theorem 7, with no deviation across the first 16 major realizations.

## 5 The Genesis Sequence

We now validate the framework by examining the sequence of structures it generates. Each realization emerges precisely when its efficiency becomes optimal relative to the dynamically rising bar, producing a progression that exhibits recognizable mathematical patterns.

### 5.1 The Complete Sequence

Table 5.1 presents the first 16 major realizations produced by the mechanized system, showing the tick $\tau$, structure realized, novelty $\nu$, effort $\kappa$, efficiency $\rho = \nu/\kappa$, and the selection bar that must be cleared.

| $\tau$ | Structure | $\nu$ | $\kappa$ | $\nu/\kappa$ | Bar $= \phi(\tau) \cdot \Omega(\tau - 1)$ |
|---|---|---|---|---|---|
| 1 | Universe $\mathcal{U}_0 : \mathcal{U}_1$ | 1 | 2 | 0.50 | — |
| 2 | Unit type $\mathbf{1}$ | 1 | 1 | 1.00 | 1.00 |
| 3 | Witness $\star : \mathbf{1}$ | 2 | 1 | 2.00 | 1.00 |
| 5 | $\Pi/\Sigma$ types | 5 | 3 | 1.67 | 1.67 |
| 8 | Circle $S^1$ | 7 | 3 | 2.33 | 2.06 |
| 13 | Propositional Truncation | 8 | 3 | 2.67 | 2.60 |
| 21 | Higher sphere $S^2$ | 10 | 3 | 3.33 | 2.98 |
| 34 | 3-Sphere $S^3 \cong \mathrm{SU}(2)$ | 18 | 5 | 3.60 | 3.44 |
| 55 | Hopf fibration $S^3 \to S^2$ | 17 | 4 | 4.25 | 4.01 |
| 89 | Generic Lie groups | 9 | 2 | 4.50 | 4.47 |
| 144 | Cohesion | 19 | 4 | 4.75 | 4.67 |
| 233 | Principal bundle connections | 26 | 5 | 5.20 | 5.06 |
| 377 | Curvature tensors | 34 | 6 | 5.67 | 5.53 |
| 610 | Metric + orthonormal frame bundle | 43 | 7 | 6.14 | 6.05 |
| 987 | Hilbert functional on Riemannian metrics | 60 | 9 | 6.67 | 6.60 |
| 1597 | Dynamical Cohesive Topos | 150 | 8 | 18.75 | 7.25 |

### 5.2 Structural Patterns

Several patterns emerge from this sequence:

### 5.2.1 Meta-Language First

The earliest realizations (R1–R5) establish the foundational type-theoretic infrastructure: universes, basic types, and dependent type formers. These structures provide maximal initial leverage—they define the language in which all subsequent mathematics will be expressed. The efficiency of $\Pi/\Sigma$ types ($\rho = 1.67$) appears modest, but these structures enable the entire polynomial functor framework, including natural numbers, products, sums, and function spaces as special cases.

### 5.2.2 The Absorption Principle

Natural numbers do not appear as a separate realization. Once $\Pi/\Sigma$ types exist, $\mathbb{N}$ becomes definable as a $W$-type:

$$\mathbb{N} \equiv W_{(b:\mathbf{2})}\text{if } b \text{ then } \mathbf{1} \text{ else } \mathbf{0}$$

Realizing $\mathbb{N}$ separately would yield efficiency $\rho \approx 1.33$, well below the bar at $\tau = 5$. The framework recognizes that dependent types provide not just natural numbers but all polynomial functors—a strictly more efficient package. This illustrates the *absorption principle*: fundamental structures are subsumed into more efficient, abstract frameworks rather than realized in isolation.

Similarly, identity types are not realized separately but bundled with the first structure that genuinely requires them—the circle $S^1$ at $\tau = 8$. The path constructor $\mathsf{loop} : \mathsf{base} =_{S^1} \mathsf{base}$ cannot be stated without identity types, so the novelty $\nu(S^1) = 7$ implicitly includes the entire path algebra machinery.

### 5.2.3 Geometric Dominance

From R5 onward, geometric structures dominate the sequence. Higher inductive types ($S^1$, $S^2$, $S^3$) appear before set-level constructions like Boolean types or finite sets. This reflects a fundamental efficiency asymmetry: geometric structures in CTT are highly relational and higher-dimensional, offering greater compression potential than discrete arithmetic structures.

The sphere sequence ($S^1 \to S^2 \to S^3$) builds systematically, with each realization enabling powerful new constructions. The identification $S^3 \cong \mathrm{SU}(2)$ at R8 is particularly significant—it's not stated as a separate theorem but emerges directly from the efficient bundling of 3-sphere structure with its natural group operations.

### 5.2.4 Progressive Abstraction

The sequence exhibits accelerating abstraction. Early realizations are concrete (specific types), middle realizations introduce frameworks (Lie groups as a general pattern), and later realizations synthesize multiple abstractions (cohesion unifies topology, logic, and geometry).

This acceleration is visible in the efficiency growth: $\rho$ increases from 0.50 (R1) to 18.75 (R16), spanning nearly two orders of magnitude. The final realization—the Dynamical Cohesive Topos—achieves exceptional efficiency by simultaneously addressing multiple mathematical frontiers (Section 7).

### 5.2.5 Fibonacci Precision

The realization times follow the Fibonacci sequence exactly: $\tau_n = F_{n+1}$. This is not approximate—across 16 major realizations, there are zero deviations from the predicted schedule. The precision is fully consistent with Theorem 7, which derives the Fibonacci schedule assuming CSH; empirically, the system behaves exactly as that conditional theorem predicts.

### 5.3 Comparison to Historical Mathematics

The Genesis Sequence is not intended as a chronological model of human mathematical history—cultures developed arithmetic long before type theory, for obvious pragmatic reasons. However, the sequence does capture something about the *logical* structure of mathematical development when viewed through the lens of efficiency.

Several historical patterns are reflected:

- **Foundational crises and resolutions**: The historical progression from naive set theory to type theory mirrors the sequence's emphasis on foundational infrastructure (R1–R5) before specific constructions
- **The geometric turn**: 20th-century mathematics increasingly emphasized geometric and topological perspectives over purely algebraic or arithmetic ones—reflected in the sequence's geometric dominance
- **Categorical abstraction**: The emergence of category theory and topos theory as unifying frameworks parallels the late-sequence appearance of cohesion (R11) and the DCT (R16)
- **Bundling related concepts**: Historically, mathematical breakthroughs often involve recognizing that apparently distinct structures share a common framework (e.g., Klein's Erlangen program). The absorption principle captures this pattern

What the sequence suggests is not that history was inevitable, but that when mathematical structures are organized by efficiency rather than historical accident, recognizable patterns emerge—patterns that may unconsciously guide mathematical intuition about which abstractions are "natural" or "fundamental."

### 5.4 Novel Predictions

The framework makes predictions beyond historical recapitulation. Most significantly, it predicts the Dynamical Cohesive Topos (R16) as a natural and highly efficient extension of geometric mathematics. While cohesive toposes and dynamical systems are well-studied separately, their systematic integration with time-equivariance and coherent modalities represents a novel synthesis.

The exceptional efficiency ($\rho = 18.75$, more than double the bar) suggests this structure addresses a genuine mathematical need—a unified framework for systems where geometric, logical, and temporal structure must evolve compatibly. Section 7 develops this construction in detail.

### 5.5 Validation and Reproducibility

The Genesis Sequence is fully reproducible. Given the axioms (Section 2), the action alphabet (primitive CTT operations with unit cost), and the mechanized implementation (Section 6), the system deterministically generates this exact sequence. The code and complete derivation certificates are available at `github.com/efficient-novelty`.

This reproducibility distinguishes the framework from post-hoc rationalization. We are not selecting structures that happen to fit a pattern; we are deriving structures from first principles and observing the patterns that emerge.

## 6 Computational Validation

The theoretical predictions of Sections 3 and 4 are validated through complete mechanization in Lean 4 [14], illustrating the power of formalization in proof assistants [4]. This section describes

how the framework was implemented, how empirical tests confirm the optimality of $k = 2$, and how the Genesis Sequence was generated.

### 6.1 Implementation Architecture

The mechanization directly encodes the formal framework from Section 2. Three core modules implement the evolutionary dynamics:

**Context representation**: The state $\mathcal{R}(\tau)$ is maintained as a cumulative record of atomic declarations (universes, type formers, constructors, eliminators, computation rules, terms), ensuring monotonicity (Axiom 1).

**Effort and novelty calculation**: Given base context $\mathcal{B}$ and horizon $H$, the system computes effort kernels $K_{\mathcal{B}}(Y)$ using iterative deepening depth-first search with memoization. Novelty is computed by comparing frontier sets $\mathcal{S}(\mathcal{B}, H)$ before and after adding candidate $X$.

**Selection mechanism**: The selection engine implements the bar mechanism (Definition **??**), candidate generation through systematic search, and winner selection according to Axiom 4.

### 6.2 Automatic Discovery of Candidates

A key feature is that candidates are discovered dynamically rather than pre-specified. For each atomic goal $Y$ not in context $\mathcal{B}$, the algorithm computes a minimal derivation under budget $H$ and extracts the delta $X = \{\text{new atoms introduced}\}$. This discovers three classes of bundles:

- Single-goal bundles (minimal sets achieving one target)
- Paired type formers (e.g., $\Pi$ and $\Sigma$ together)
- Complete higher inductive types (type former + constructors + eliminator)

The system enforces level discipline: candidates at level $L^*$ can only use prerequisites from levels $L^* - 1$ or $L^*$, preventing circularity.

### 6.3 Empirical Validation of $k = 2$ Optimality

The implementation allows testing alternative integration strategies by constraining which layers are accessible during sealing.

**Testing** $k = 1$: When restricted to $\mathcal{S}_1$ (only accessing $L_n$), the system stalls at $\Pi/\Sigma$ types. The sealing process cannot construct required 2D witnesses (e.g., for the Interchange Law) because it lacks access to $L_{n-1}$. The candidate is inadmissible, confirming Theorem 4.

**Testing** $k = 3$: When both $\mathcal{S}_2$ and $\mathcal{S}_3$ implementations are available for each candidate, the selection criterion consistently chooses $\mathcal{S}_2$ variants. For the circle $S^1$:

- $\mathcal{S}_2$: $\kappa^{(2)} = 3$, $\nu^{(2)} = 7$, $\rho^{(2)} = 2.33$
- $\mathcal{S}_3$: $\kappa^{(3)} = 4$, $\nu^{(3)} = 7$, $\rho^{(3)} = 1.75$

The redundant checks against $L_{n-2}$ increase effort without enabling new constructions, exactly as predicted by Theorem 6.

### 6.4 Genesis Sequence Generation

Running the system with optimal $\mathcal{S}_2$ integration generates the sequence presented in Table 5.1. Key validation points:

**Fibonacci timing**: Realization times satisfy $\tau_n = F_{n+1}$ exactly across all 16 major realizations, with zero deviations, matching the conditional prediction of Theorem 7 under CSH.

**Efficiency growth**: Realized efficiencies increase from $\rho_1 = 0.50$ to $\rho_{16} = 18.75$, demonstrating unbounded growth as predicted by Theorem 1.

**Absorption patterns**: Natural numbers and identity types do not appear as separate realizations, confirming the absorption principle (Section 5).

**Geometric dominance**: Higher inductive types dominate the middle sequence, consistent with the efficiency of relational, higher-dimensional structures.

### 6.5 Reproducibility and Auditing

The implementation is fully deterministic. Given the axioms and action alphabet (unit-cost primitive CTT operations), it generates identical sequences across runs. The code includes comprehensive certificate generation:

- `PackageCert`: validates effort calculations
- `FrontierCert`: confirms individual frontier entries
- `NoveltyCert`: bundles complete novelty computations

Each certificate carries witnesses enabling independent verification of both derivation correctness and novelty accounting. Complete source code, documentation, and test suites are available at `github.com/efficient-novelty`.

The mechanization processes approximately 100-200 candidates per tick at later stages, with memoization and pruning strategies keeping computation tractable despite the combinatorial search space.

## 7 The Dynamical Cohesive Topos

At $\tau = 1597$, the framework generates a structure achieving exceptional efficiency ($\rho = 18.75$, more than double the selection bar of 8.29). We call this construction the *Dynamical Cohesive Topos* (DCT)—a formalization of how time actions interact coherently with geometric and logical structure.

While DCTs build on established concepts—particularly Schreiber's cohesive $(\infty, 1)$-toposes [9], Lawvere's cohesion in topos theory, and the Joyal–Tierney axiomatization of toposes [13, 12] alongside classical dynamical systems theory—they represent a systematic integration: axiomatizing how temporal evolution preserves cohesive modalities and studying the resulting categorical structure. Related cohesive-dynamical perspectives appear in temporal/cohesive topos models such as Schultz and Spivak's temporal type theory [21], where time-indexed geometric morphisms are required to commute with the cohesive adjunctions. DCTs isolate these compatibilities abstractly: the equivariance data packages the Beck–Chevalley conditions needed in cohesive $(\infty, 1)$-toposes while allowing general monoidal time actions.

The resulting temporal modality $\triangleright$ and cohesion-equivariance axioms echo the guarded and Fitch-style modalities used to control recursion and coinduction. Nakano's "later" operator $\blacktriangleright$ [24] and its guarded generalizations in dependent and cubical type theory [23, 22] impose naturality conditions matching the $\alpha_t^{\mathsf{Shape}}$, $\alpha_t^{\flat}$, and $\alpha_t^{\sharp}$ coherence data. Temporal Type Theory internalizes a similar interaction between cohesive structure and time-shift modalities [21]. DCTs thus position the PEN-generated structure squarely within the existing landscape of cohesive $(\infty, 1)$-toposes equipped with guarded/temporal modalities, but present it as a single axiomatized package tailored for evolutionary dynamics.

### 7.1 Definition and Axioms

Throughout, let $(T, \oplus, 0)$ denote a monoid object interpreted as "time" (e.g., $\mathbb{N}$ for discrete time or $\mathbb{R}_{\geq 0}$ for continuous time). Write $\mathsf{GeomEnd}(T)$ for the $(\infty, 1)$-category of geometric endomorphisms of an $(\infty, 1)$-topos $\mathcal{T}$.

**Definition 12** (Dynamical Cohesive Topos)**.** A Dynamical Cohesive Topos is a tuple $\mathcal{D} = (\mathcal{T}, \mathsf{coh}, \mathcal{A})$ consisting of:

1. An $(\infty, 1)$-topos $\mathcal{T}$ [10]

2. A cohesive structure over a base $(\infty, 1)$-topos $\mathcal{S}$:

$$\Pi \dashv \mathsf{Disc} \dashv \Gamma \dashv \mathsf{Codisc} : \mathcal{T} \rightleftarrows \mathcal{S}$$

   satisfying the usual axioms of cohesion. Equivalently, we work with the adjoint triple $\mathsf{Shape} \dashv \flat \dashv \sharp$ relative to $\mathcal{S}$.

3. A strong monoidal action of time by geometric endomorphisms:

$$\mathcal{A} : (T, \oplus, 0) \to \mathsf{GeomEnd}(\mathcal{T}), \quad t \mapsto \mathcal{A}_t^* : \mathcal{T} \to \mathcal{T}$$

   equipped with coherent isomorphisms $\mathcal{A}_0^* \simeq \mathsf{id}_{\mathcal{T}}$ and $\mathcal{A}_{s\oplus t}^* \simeq \mathcal{A}_t^* \circ \mathcal{A}_s^*$, where each $\mathcal{A}_t^*$ is left exact and accessible (as an inverse-image functor of a geometric morphism).

These data satisfy *cohesion-equivariance axioms*: for each $t \in T$ there are specified invertible 2-cells, natural in $X \in \mathcal{T}$:

$$\begin{aligned} \alpha_t^{\mathsf{Shape}} &: \mathsf{Shape}(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\mathsf{Shape}\, X) \\ \alpha_t^{\flat} &: \flat(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\flat X) \\ \alpha_t^{\sharp} &: \sharp(\mathcal{A}_t^* X) \xrightarrow{\sim} \mathcal{A}_t^*(\sharp X) \end{aligned}$$

which satisfy monoidal coherence with respect to 0 and $\oplus$ and are compatible with unit/counit data of the cohesive adjunctions (Beck-Chevalley/Frobenius conditions).

If $\mathcal{T}$ carries differential cohesion, a DCT requires that $\mathcal{A}$ preserves the differential structure via analogous comparison 2-cells.

*Remark* 6 (Flows on Sites). Many concrete DCTs arise when a time action on a site induces pullback along "flow maps." For example, if a space $X$ carries $a : T \times X \to X$, then $\mathsf{Sh}(X)$ inherits $\mathcal{A}_t^* := a_t^{-1}$; the cohesion is standard relative to spaces, and comparison maps follow from naturality of adjoints.

## 7.2 Morphisms and the Category DCT

**Definition 13** (Morphisms of DCTs)**.** Let $\mathcal{D}_i = (\mathcal{T}_i, \mathsf{coh}_i, \mathcal{A}^i)$ for $i = 1, 2$. A DCT-morphism $F : \mathcal{D}_1 \to \mathcal{D}_2$ consists of:

1. A geometric morphism $F : \mathcal{T}_1 \rightleftarrows \mathcal{T}_2$ with inverse image $F^* : \mathcal{T}_2 \to \mathcal{T}_1$ preserving cohesive structure (canonical comparison 2-cells $F^*\mathsf{Shape}_2 \Rightarrow \mathsf{Shape}_1 F^*$, etc., are invertible)

2. A coherent $T$-equivariance structure:

$$\vartheta_t : F^* \circ \mathcal{A}_t^{2*} \xrightarrow{\sim} \mathcal{A}_t^{1*} \circ F^*$$

   which is monoidal in $t$ and compatible with cohesion-equivariance 2-cells

**Proposition 1** (Category Structure)**.** *DCTs and their morphisms form an $(\infty, 1)$-category* **DCT**. *Finite products exist: given $\mathcal{D}_1, \mathcal{D}_2$, the product has underlying topos $\mathcal{T}_1 \times \mathcal{T}_2$, product cohesion, and componentwise action $t \mapsto (\mathcal{A}_t^{1*}, \mathcal{A}_t^{2*})$.*

### 7.3 Examples

*Example* 1 (Rotation on a Circle). Let $X = S^1$ with $a : \mathbb{R}_{\geq 0} \times S^1 \to S^1$, $a(t, x) = x + \alpha t$ for fixed $\alpha$. Set $\mathcal{T} = \mathsf{Sh}(S^1)$ with cohesion relative to spaces, and $\mathcal{A}_t^* = a_t^{-1}$. The action is strong monoidal, each $\mathcal{A}_t^*$ is geometric, and cohesion-equivariance maps follow from naturality.

*Example* 2 (Scaling Flow on Euclidean Space). Let $X = \mathbb{R}^n$ with $a(t, x) = e^{Ht}x$ $(H \geq 0)$. Then $\mathcal{T} = \mathsf{Sh}(\mathbb{R}^n)$ and $\mathcal{A}_t^* = a_t^{-1}$ define a DCT modeling an "expanding universe" where geometric and logical structure evolve compatibly.

*Example* 3 (Discrete Time via Presheaves). Let $\mathcal{C}$ be a small category with endofunctor $U : \mathcal{C} \to \mathcal{C}$. Put $\mathcal{T} = \widehat{\mathcal{C}} = \mathsf{Psh}(\mathcal{C})$, time $T = (\mathbb{N}, +, 0)$, and $\mathcal{A}_n^* := (-) \circ U^n$. If $\mathcal{T}$ has (relative) cohesive structure, the action is left exact and cohesion-equivariance holds when adjoints are defined objectwise.

*Example* 4 (Smooth $\infty$-Groupoids). Let $\mathcal{T} = \mathsf{Sh}_\infty(\mathsf{Man})$ with differential cohesion. A smooth flow $\phi : \mathbb{R}_{\geq 0} \times M \to M$ induces an action on $\mathsf{Sh}_\infty(M)$ by pullback $\mathcal{A}_t^* = \phi_t^*$. The differential cohesive structure is preserved by diffeomorphisms, yielding a differential DCT.

### 7.4 Structural Properties

**Proposition 2** (Invariants as Reflective Subtopos)**.** *Assume $\mathcal{A}_t^*$ preserves all small limits. The full subcategory*

$$\mathcal{T}^{\mathcal{A}} = \left\{ X \in \mathcal{T} \mid \theta_t : \mathcal{A}_t^* X \xrightarrow{\sim} X \text{ for all } t \in T, \text{ coherently} \right\}$$

*of $\mathcal{A}$-invariant objects is an $(\infty, 1)$-subtopos. The inclusion $i : \mathcal{T}^{\mathcal{A}} \hookrightarrow \mathcal{T}$ admits a left exact left adjoint (reflector) computed as the limit of the action diagram.*

*If cohesion-equivariance maps are invertible, then* $\mathsf{Shape}$, $\flat$, *and* $\sharp$ *restrict to $\mathcal{T}^{\mathcal{A}}$ and commute with $i$ up to isomorphism. In particular:*

$$\mathcal{A}_t^*(\flat X) \simeq \flat X, \quad \mathcal{A}_t^*(\sharp X) \simeq \sharp X$$

*The "discrete" and "codiscrete" parts are conserved—a categorical formalization of conservation laws.*

**Proposition 3** (Temporal Modality)**.** *Suppose $T$ contains a cofinal "future" submonoid $T_{>0} \subseteq T$, and $\mathcal{T}$ has small limits preserved by each $\mathcal{A}_t^*$ and by the cohesive adjoints. Then the endofunctor*

$$\triangleright X := \lim_{t \in T_{>0}} \mathcal{A}_t^* X$$

*exists and carries canonical comparison isomorphisms:*

$$\mathsf{Shape}(\triangleright X) \simeq \triangleright(\mathsf{Shape}\, X), \quad \flat(\triangleright X) \simeq \triangleright(\flat X), \quad \sharp(\triangleright X) \simeq \triangleright(\sharp X)$$

*The unit $X \to \triangleright X$ (projection to the $t > 0$ limit) serves as a "later"/guarded modality, useful for internal temporal reasoning.*

*Remark* 7 (Guarded semantics). The endofunctor $\triangleright$ reproduces Nakano's later modality and its guarded dependent generalizations when interpreted in the internal language of a DCT. The cofinal limit implements the clock-synchronization mechanism used in guarded dependent type theory with clocks [23] and in guarded cubical type theory [22], while the equivariance 2-cells ensure that guarded rules commute with the cohesive adjoints, mirroring the modal laws isolated in Temporal Type Theory [21]. Consequently, propositions proven using DCT's $\triangleright$ translate directly to the guarded recursion principles familiar from these settings.

### 7.5 Why This Structure Emerges

The DCT achieves exceptional efficiency ($\rho = 18.75$) by simultaneously addressing multiple mathematical frontiers that had developed independently in the Genesis Sequence:

**Geometric structure** (cohesion): Realized at R11, providing the Shape/$\flat$/$\sharp$ modalities that distinguish homotopy types, discrete structures, and codiscrete structures.

**Dynamical structure** (Lie groups, flows): Emergent through R8–R10, capturing continuous symmetry and transformation.

**Logical structure** (type theory, modalities): Present from R1–R5, providing the foundational language.

The DCT synthesizes these by axiomatizing their coherent interaction: time actions preserve geometric distinctions, conservation laws emerge naturally (Proposition 2), and temporal reasoning integrates with spatial reasoning (Proposition 3).

The high novelty ($\nu = 150$) reflects that DCTs enable constructions across all three domains simultaneously:

- Differential invariants under flows (geometric dynamics)
- Temporal logic with spatial modalities (logical dynamics)
- Classification of equivariant structures (categorical dynamics)

The modest effort ($\kappa = 8$) reflects that the axiomatization is clean: three components (topos, cohesion, action) with three coherence conditions (equivariance 2-cells), all fitting naturally within CTT's infrastructure.

### 7.6 Open Directions

Several natural questions emerge from this construction:

**Differential dynamics**: In a differential DCT, under what conditions does $\mathcal{A}$ preserve connections or send geodesics to geodesics? Proposition 2 identifies differential invariants.

**Classification**: For fixed cohesive $\mathcal{T}$, classify time actions up to monoidal natural equivalence. Study moduli of actions with prescribed invariants.

**Internal type theory**: The modality $\triangleright$ (Proposition 3) underlies a temporal/guarded homotopy type theory internal to a DCT. Equivariance ensures modal rules commute with Shape, $\flat$, $\sharp$.

**Exactness properties**: Equivariant geometric morphisms form an $(\infty, 1)$-category with (finite) limits. Exploring exactness and stability properties is natural.

The emergence of DCTs at $\tau = 1597$ suggests this synthesis addresses a genuine mathematical need—a prediction the framework makes about productive future directions in geometric type theory.

## 8 The Central Open Question: From Structure to Dynamics

We have proven that two-layer integration is optimal for CTT (Section 4). We have observed that the mechanized system exhibits Fibonacci timing with zero deviation (Section 5). The Complexity Scaling Hypothesis bridges these results, but remains unproven.

### 8.1 What We Know

**Proven:** CTT's coherence structure is fundamentally 2-dimensional (Theorem 5)

**Proven:** Two-layer integration is necessary and sufficient (Theorems 4, 6)

**Observed:** Integration complexity follows $\Delta_n = F_n$ exactly for $n \geq 4$

**Conjectured:** This arises from additive scaling of coherence obligations (CSH)

### 8.2 Interpreting the Complexity Scaling Hypothesis

CSH claims that, beyond a short bootstrap phase, the interface basis behaves like a two-dimensional boundary: every new realization interacts independently with the two most recent layers, and the cross-layer witnesses introduced during this interaction stabilise immediately. This remains a conjectural structural statement about CTT. It says that once a coherence schema has been witnessed for adjacent layers, no higher-order correction is needed when later structures refer back to the same schema.

Two ingredients justify treating CSH as a plausible hypothesis rather than a mere empirical curiosity:

1. *Dimensional saturation*: Theorems 4 and 5 show that CTT's coherence laws are intrinsically two-dimensional. When a new definition references $L_n$ and $L_{n-1}$, it cannot generate genuinely new three-way overlaps because such overlaps would require simultaneous access to $L_{n-2}$, which is excluded by optimal integration.

2. *Interface reuse*: Each schema in $\mathcal{I}_n$ represents a normal form for a coherence witness. When the same schema reappears in $\mathcal{I}_{n+1}$ the witness constructed at stage $n$ can be transported forward using the Kan operations already available. No new definitional work is needed beyond a constant amount of bookkeeping.

Taken together, these points suggest that interface complexity is dominated by the additive contributions from $L_n$ and $L_{n-1}$, with cross-terms bounded. Yet they fall short of a proof: the bounded error $\varepsilon_{n+1}$ in Hypothesis 1 formalizes a saturation principle that still awaits derivation from the axioms of CTT.

### 8.3 Evidence from the Genesis Sequence

The Lean implementation provides concrete diagnostics supporting CSH:

- **Witness counts**: Certificate logs record the number of sealing obligations discharged at each realization. From R5 onward these counts satisfy $\sigma_{n+1} = \sigma_n + \sigma_{n-1}$ exactly, with $\sigma_1 = \sigma_2 = 1$.

- **Stability of cross-terms**: When a schema such as the Interchange Law appears in consecutive layers, later uses do not spawn additional witnesses—only transport compositions, which the implementation normalises for free. This corroborates the stabilization clause of Hypothesis 1.

- **Robustness to perturbations**: We experimented with alternative unit-cost assignments (weighing transport versus composition differently). While absolute efficiencies shift, the recurrence of witness counts remains intact provided the costs preserve additivity over disjoint unions, indicating that the pattern is structural rather than numerical coincidence.

The numerical equality between certificate counts and Fibonacci numbers therefore has both empirical and conceptual backing. CSH distills this into a testable assertion about the geometry of coherent integration.

### 8.4 Interrogating the Hypothesis

Validating or refuting CSH demands a three-pronged programme, each designed to surface potential counterexamples as readily as supporting evidence:

**Fine-grained audits** Enrich the mechanization with tracing tools that record how often specific schemas require new witnesses versus transports. Showing that only $O(1)$ genuinely new witnesses appear after bootstrap would tighten the empirical case.

**Semantic analysis** Translate Hypothesis 1 into the internal language of CTT: characterize interface schemas as generators of a 2-truncated coherence operad and prove that its bar construction terminates after two steps. Such a result would elevate CSH from hypothesis to theorem.

**Adversarial tests** Force counterexamples by modifying the action alphabet to include intentionally difficult coherence conditions (e.g., exotic higher inductive types) and observe whether the hypothesis fails. Identifying the boundary of validity clarifies the structural assumptions being made.

These investigations would either solidify CSH or expose where additional structure beyond two-layer integration is necessary, furnishing a clear path to falsification if the hypothesis fails.

### 8.5 Beyond Cubical Type Theory

Although proven in the CTT setting, CSH hints at a more universal phenomenon: efficient mathematical evolution may be constrained by how coherence information scales. We therefore retain and refine our broader conjecture.

**Conjecture 1** (Cross-System Complexity Scaling)**.** *Let $\{F_i\}$ be Turing-complete formal systems equipped with cumulative libraries and a notion of effort/novelty analogous to Section 2. Suppose a discovery trajectory in $F_i$ respects a two-layer integration policy. Then there exists a partial order $\preceq$ on mathematical concepts such that:*

1. *Each trajectory eventually respects $\preceq$ up to incomparabilities.*
2. *The integration gaps in any such trajectory satisfy a linear recurrence whose dominant root exceeds 1; in systems admitting CSH this recurrence is Fibonacci.*
3. *Deviations from the Fibonacci law witness explicit violations of CSH (e.g., non-stabilizing cross-layer witnesses).*

Conjecture 1 generalizes the Turing universality perspective: it predicts a common skeleton for efficiency-driven discovery while allowing the dominant recurrence to vary with how interface complexity scales. Demonstrating or refuting the existence of $\preceq$ requires the same information-theoretic translation sketched earlier, now sharpened around CSH.

### 8.6 Experimental and Theoretical Next Steps

The most promising avenues for future work align with this conjectural picture:

- **Cross-foundational replications**: Mechanize PEN-style evolution in ZFC, the Calculus of Inductive Constructions, and elementary topos theory. Measure whether interface growth is additive (supporting CSH) or exhibits different scaling (suggesting other linear recurrences).
- **Complexity-theoretic grounding**: Relate $\kappa$ to Kolmogorov complexity as in the discussion preceding Hypothesis 1. Showing that CSH corresponds to subadditivity of conditional Kolmogorov complexity would give it machine-independent semantics.

- **Geometric versus arithmetic regimes**: Extend the absorption analysis of Section 5 by comparing compression ratios for geometric and arithmetic structures. If CSH holds only in the geometric regime, this would explain the observed bias and set expectations for where Fibonacci timing should break down.

Each of these steps directly interrogates the mechanism behind the golden-ratio schedule rather than the schedule itself.

### 8.7 Implications for Mathematical Practice

Should CSH withstand scrutiny, it offers a concrete explanation for why certain abstractions feel "inevitable": they arise precisely when the available coherence boundary can absorb them with Fibonacci efficiency. Conversely, constructing contexts where CSH fails would illuminate genuinely new frontiers—places where mathematicians should expect surprises because the usual scaling intuition breaks.

Even if the hypothesis is ultimately false in other foundations, its very failure would be informative. It would isolate which aspects of CTT (higher-dimensional identity, Kan operations, modality interactions) enforce the scaling law, guiding both foundational comparisons and practical proof-engineering strategies.

In summary, the Complexity Scaling Hypothesis is the hinge on which the Genesis Sequence's Fibonacci rhythm turns. Exploring its mathematical content—proving it, refuting it, or delineating its domain of validity—constitutes the central agenda for extending the Principle of Efficient Novelty beyond the specific model analysed here.

## 9 Conclusion

We opened with a question: why do some abstractions immediately reorganize mathematical thinking while others remain curiosities? This paper provides a partial answer.

Within Cubical Type Theory, we have proven that mathematical structures emerging under efficiency optimization follow precise laws. Conditional on the Complexity Scaling Hypothesis, the Fibonacci timing is not a statistical tendency or modeling artifact—it becomes a mathematical theorem about CTT's geometry (Theorem 7), arising from the two-dimensional nature of its coherence requirements (Theorems 5, 6). The mechanized Genesis Sequence validates these predictions with zero deviations across 16 major realizations, lending support to the hypothesis.

Whether similar patterns emerge in fundamentally different foundational systems remains an open question (Conjecture 1). The path forward requires implementing analogous frameworks in ZFC, category theory, and other foundations to test whether efficiency-driven discovery converges to stable orderings across systems.

But even confined to CTT, our results establish something significant: type-theoretic foundations possess deep structural properties constraining how mathematical knowledge can be efficiently organized, echoing Simon's analysis of hierarchical architectures in complex systems [19]. The feeling that $\pi_1(S^1) \simeq \mathbb{Z}$ is more profound than defining another arbitrary group may reflect our implicit recognition of exceptional efficiency—a cognitive response to discovering maximally compressive abstractions.

Mathematics may not be arbitrary. At minimum, within the formal universe of cubical type theory, its progression follows laws as precise as those governing physical systems. We have proven that two-layer integration is optimal; we have observed that this produces Fibonacci timing with extraordinary precision.

Whether the Complexity Scaling Hypothesis—the bridge between these results—is a theorem waiting to be proven or an artifact of our specific implementation remains the central open question. Resolving it will require either:

- A semantic proof that 2D coherence entails additive interface scaling, or
- Discovery of counterexamples where optimal integration produces different recurrence relations

Either outcome would deepen our understanding of how foundational structure constrains mathematical possibility. The shape of that constraint—whether Fibonacci or otherwise—awaits rigorous determination.

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